

# Master Study Guide: Engineering Physics (Mechanics and Optics)

## Unit 1: Elasticity — [16% - 27%] of Total Marks

### A. Short Answer Questions (SAQs)

\*\*\*2082 Mid-Term Q1 (5):\*\*\*

Define bending moment. Derive its relation for a beam. [1+4]

\*\*\*2082 Mid-Term Q3 (5):\*\*\*

Two cylinders have the same length and mass and are made of same materials; one is solid while the other which is hollow has an external radius twice the internal radius. Compare the torsional rigidities of the two cylinders. Show that second is stronger. **[REPEATED: Similar concept to 2080 Q3(b)]**

\*\*\*2079 Q6 (8):\*\*\*

Compare the loads required to produce equal depressions for two beams made of same material and having the same length and weight with only difference that one has circular cross-section while the cross-section of the other is square.

\*\*\*2080 Q3 (8):\*\*\*

- **(a) [4 Marks]:** Define Young's modulus, modulus of rigidity and Poisson's ratio, and write relation between them. **[REPEATED: Similar to 2079 Q7(a), 2081 Q7(a)]**
- **(b) [4 Marks]:** Explain why a hollow cylinder is stronger than solid cylinder of the same length, mass and material. **[REPEATED: Similar to 2082 Mid-Term Q3]**

**2080 Q4(b) (4):**

What is elasticity? Discuss elastic hysteresis. [1+3]

### B. Long Answer Questions (LAQs)

\*\*\*2081 Q7 (15):\*\*\*

- **(a) [10 Marks]:** Define the term stress and strain. If  $Y$ ,  $K$  and  $\sigma$  represents Young's modulus, Bulk modulus and Poisson's ratio respectively, then prove that  $K = Y / [3(1 - 2\sigma)]$ . [2+8] **[REPEATED: Identical to 2081 Mid-Term Q5(a), 2079 Q7(a)]**
- **(b) [5 Marks]:** What couple must be applied to a wire one meter long, 1 mm in diameter in order to twist one end of it, through 90 degrees, the other end remaining fixed. Rigidity of material of the wire is  $2.8 \times 10^{10} \text{ Nm}^{-2}$ . **[REPEATED: Identical to 2079 Q7(b)]**

**2081 Q11(b) (5):**

A brass bar  $1\text{ cm}^2$  in cross-section is supported on two knife edges one meter apart. A load  $1\text{ kg}$  at the center of the bar depresses that point  $2.51\text{ mm}$ . What is Young's modulus of brass.

**2081 Mid-Term Q3 (8):**

What is cantilever? Derive an expression for the depression of the loaded end of cantilever having rectangular cross-section of negligible weight. **[REPEATED: Similar derivation to 2080 Q7(a)]**

**2081 Mid-Term Q5(a) (11):**

Define Young's modulus, Bulk modulus and Poisson's ratio. If  $Y$ ,  $K$  and  $\sigma$  represent Young's modulus, Bulk modulus and Poisson's ratio respectively, prove that  $K = Y / [3(1 - 2\sigma)]$ . **[3+8]**  
**[REPEATED: Identical to 2081 Q7(a), 2079 Q7(a)]**

**2080 Q7(a) (10):**

What is cantilever? Derive an expression for the depression of loaded end of a cantilever of circular cross-section of negligible weight. **[2+8]** **[REPEATED: Similar derivation to 2081 Mid-Term Q3]**

**2079 Q7 (15):**

- **(a) [10 Marks]:** Define Young's modulus ( $Y$ ), Bulk modulus ( $K$ ) and Poisson's ratio ( $\sigma$ ). Derive the relation between them. **[REPEATED: Identical to 2081 Q7(a)]**
- **(b) [5 Marks]:** What couple must be applied to a wire  $1\text{ m}$  long,  $1\text{ mm}$  in diameter in order to twist one end of it, through  $90^\circ$ , the other end remaining fixed. Modulus of rigidity of material of the wire is  $2.8 \times 10^{10}\text{ N/m}^2$ . **[REPEATED: Identical to 2081 Q7(b)]**

**2079 Q10(a) (10):**

Define neutral surface and bending moment of a beam. Deduce the relation for bending moment of a beam. **[REPEATED: Identical derivation to 2082 Mid-Term Q1]**

**C. Trend Analysis & Focal Areas**

- **Question Types & Styles:** This unit is heavily focused on theoretical derivations of elastic relations and bending parameters. Numerical questions consistently test torsional couples on wires or depressions in supported beams/cantilevers.
- **Direct Repetitions:**
  - The relation proof  $K = Y / [3(1 - 2\sigma)]$  appears in 2079, 2080, 2081, and 2081 Mid-Term.
  - The wire twist couple problem using  $2.8 \times 10^{10}\text{ N/m}^2$  is identical in 2079 and 2081.

- **Key High-Priority Topics:**

- *Proof of relation between  $Y$ ,  $K$ ,  $\eta$ , and  $\sigma$ .*
- *Bending moment of a beam and Cantilever depressions (circular and rectangular cross-sections).*
- *Torsional rigidity comparison of solid vs hollow cylinders.*

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## **Unit 2: Fluid Dynamics & Viscosity — [15% - 19%] of Total Marks**

### **A. Short Answer Questions (SAQs)**

\*\*\*2079 Q1 (8):\*\*\*

State Bernoulli's theorem. Show how Bernoulli's theorem is applied to measure the rate of discharge through the city water main's.

\*\*\*2081 Q4 (8):\*\*\*

State assumptions made and with necessary theory, explain Poiseuille's method of determining the coefficient of viscosity of a liquid. **[REPEATED: Identical to 2080 Q8(a)]**

\*\*\*2080 Q4(a) (4):\*\*\*

What is Reynold number? Explain significance of Reynold's number. **[1+3]**

\*\*\*2080 Q5(a) (4):\*\*\*

Derive equation of continuity for steady flow of liquid.

### **B. Long Answer Questions (LAQs)**

\*\*\*2081 Q8 (15):\*\*\*

- **(a) [10 Marks]:** State and prove Bernoulli's theorem for a liquid along a stream line.
- **(b) [5 Marks]:** A water drop of radius 0.01 cm is falling through air. Find its terminal velocity. Neglect the density of air. (coefficient of viscosity of air is  $1.8 \times 10^{-4}$  cgs unit)  
**[REPEATED: Similar Stokes' Law concept to 2081 Mid-Term Q5(b)]**

**2081 Mid-Term Q4 (8):**

Define stream line flow. Derive Poiseuille's equation. **[2+6]**

**2081 Mid-Term Q5(b) (5):**

Find the terminal velocity of a free falling water drops of radius 0.04 mm. The coefficient of viscosity of air is  $1.9 \times 10^{-5}$  Ns/m<sup>2</sup> and its density is **1.2 Kg/m<sup>3</sup>**, density of water is **1000 Kg/m<sup>3</sup>**.  
**[REPEATED: Similar Stokes' Law concept to 2081 Q8(b)]**

**2080 Q7(b) (5):**

A pipe is running full of water. At a certain point A it tapers from 0.6m diameter to 0.2m diameter at B. The pressure difference between A and B is 1m of water column. Find rate of flow of water through the pipe.

**2080 Q8(a) (10):**

Discuss Poiseuille's assumptions and derive Poiseuille's formula for the rate of flow of a liquid through tube. [2+8] [REPEATED: Identical to 2081 Q4]

**2079 Q8(b) (5):**

A water drop is observed to fall through a gas of density 0.001 gm/cc with a constant velocity of 98 cm/s. What is the radius of the drop? The coefficient of viscosity of the gas is  $2 \times 10^{-4}$  poise.

**2079 Q10(b) (5):**

Capillary tube of length  $l$ ,  $2l$  and  $l/2$  are connected in series, the radii being  $r$ ,  $r/2$  and  $r/3$  respectively. If a streamline flow is maintained and the pressure across first capillary is  $P_1$ , deduce pressure across the second and third capillaries.

**C. Trend Analysis & Focal Areas**

- **Question Types & Styles:** Focuses on derivations of fluid theorems (Bernoulli, Continuity, Poiseuille) and numericals targeting terminal velocity of falling drops (Stokes' law application) or pipe/capillary flow rates.
  - **Direct Repetitions:**
    - Poiseuille's equation derivation/viscosity determination method in 2080, 2081, and 2081 Mid-Term.
    - Stokes' Law terminal velocity calculation (falling water drop) in 2079, 2081, and 2081 Mid-Term.
  - **Key High-Priority Topics:**
    - Derivation of Bernoulli's theorem and Continuity equation.
    - Poiseuille's equation derivation and series connection problems.
    - Stokes' Law terminal velocity numericals.
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## **Unit 3: Surface Tension — [7% - 8%] of Total Marks**

### **A. Short Answer Questions (SAQs)**

\*\*\*2080 Q5(b) (4):\*\*\*

What is surface tension? Explain why a liquid is either raised or depressed in the capillary tube?  
[1+3]

### **\*\*B. Long Answer Questions (LAQs)\*\***

\*\*\*2082 Mid-Term Q5(b) (3):\*\*\*

A 0.02 cm liquid column balances the excess pressure inside a soap bubble of radius 7.5 mm. Determine the density of the liquid. Surface tension of soap solution is  $0.03 \text{ Nm}^{-1}$ .

\*\*\*2081 Q11(a) (10):\*\*\*

Define surface tension. Deduce an expression for the difference of pressure on the two sides of a spherical drop. [2+8] **[REPEATED: Identical to 2079 Q8(a)]**

\*\*\*2080 Q8(b) (5):\*\*\*

Calculate the work done in spraying a drop of mercury of 1 mm radius into one million identical drops all of the same size. (Surface tension of mercury is  $550 \times 10^{-3} \text{ N/m}$ )

\*\*\*2079 Q8(a) (10):\*\*\*

Define surface tension and angle of contact. Determine the expression for excess pressure inside a drop. **[REPEATED: Identical to 2081 Q11(a)]**

### **C. Trend Analysis & Focal Areas**

- **Question Types & Styles:** *This is a compact, high-yielding chapter. Numericals feature work done in droplet spraying or excess pressure calculations, while theory asks for excess pressure derivations.*
  - **Direct Repetitions:**
    - *Excess pressure inside a spherical drop derivation in 2079 and 2081.*
  - **Key High-Priority Topics:**
    - *Derivation of excess pressure inside a liquid drop / soap bubble.*
    - *Work done in splitting/spraying liquid drops (droplet surface energy problems).*
    - *Capillary rise/depression phenomenon.*
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## **Unit 4: Wave Optics: Interference — [11% - 19%] of Total Marks**

### **A. Short Answer Questions (SAQs)**

\*\*\*2082 Mid-Term Q2 (5):\*\*\*

Define interference of light. Explain the working of Fresnel biprism. [1+4] [REPEATED: Similar theory to 2081 SAQ3, 2080 Q10(a)]

\*\*\*2081 Q3 (8):\*\*\*

Describe Fresnel's biprism and explain in detail how the wavelength of monochromatic source of light can be determined by using the biprism. [REPEATED: Identical to 2080 Q10(a)]

\*\*\*2080 Q1 (8):\*\*\*

- **(a) [4 Marks]:** Define phase of a particle. Give the relation between phase difference and path difference. [2+2]
- **(b) [4 Marks]:** What is interference? What will happen in interference fringe in double slit experiment if one slit is covered? [1+3]

**2079 Q3 (8):**

Define interference of light and coherent sources. Explain the techniques for creating coherent sources of light. [REPEATED: Similar to 2082 Mid-Term Q6(a)]

### **B. Long Answer Questions (LAQs)**

\*\*\*2081 Q9(b) (5):\*\*\*

A Lloyd's mirror of length 5 cm is illuminated by light of wavelength 600 nm from a narrow slit 0.1 cm above the plane of mirror. Calculate the fringe width on a screen 120 cm from the slit.

\*\*\*2081 Mid-Term Q1 (8):\*\*\*

Describe the principle and working of Fabry-Perot interferometer and determine its intensity.

\*\*\*2081 Mid-Term Q2 (8):\*\*\*

Derive an expression to determine fringe width of wedge-shaped film.

\*\*\*2080 Q10 (15):\*\*\*

- **(a) [10 Marks]:** What is Fresnel's biprism? Describe necessary theory to determine wavelength of light by using Fresnel biprism. [1+9] [REPEATED: Identical to 2081 Q3]
- **(b) [5 Marks]:** Two slits are 0.3 mm apart and placed 50cm from screen. What is distance between the second and third dark line of the interference pattern when the slits are illuminated with a light of 600nm wavelength.

**2079 Q9(a) (10):**

Describe Michelson interferometer with diagram. Explain how the wavelength of light can be determined with its help.

**C. Trend Analysis & Focal Areas**

- **Question Types & Styles:** *Practical setups of interference (Fresnel Biprism, Lloyd's Mirror, Michelson & Fabry-Perot Interferometers) form the backbone. Calculations deal with fringe widths and slit distances.*
  - **Direct Repetitions:**
    - *Fresnel Biprism theory and wavelength determination is a consistent staple in 2080, 2081, and 2082 Mid-Term.*
  - **Key High-Priority Topics:**
    - *Fresnel Biprism setup and fringe width theory.*
    - *Michelson Interferometer & wedge-shaped thin film equations.*
    - *Coherent source creation methods.*
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**Unit 5: Wave Optics: Diffraction — [15% - 21%] of Total Marks**

**A. Short Answer Questions (SAQs)**

\*\*\*2082 Mid-Term Q4 (5):\*\*\*

Differentiate between Fraunhofer and Fresnel diffraction. Discuss Fraunhofer single slit diffraction. [1+4] [REPEATED: Similar to 2079 Q11(a)]

\*\*\*2081 Q2 (8):\*\*\*

What is a Zone plate? Derive an expression for its focal length. [2+6] [REPEATED: Similar to 2080 Q9(a)]

\*\*\*2081 Q5 (8):\*\*\*

Define resolving power. Derive an expression for resolving power of plane transmission grating. [2+6] [REPEATED: Identical to 2079 Q5]

\*\*\*2080 Q6(b) (4):\*\*\*

Differentiate between diffraction and interference of light.

\*\*\*2079 Q5 (8):\*\*\*

Define resolving power. Derive the expression for resolving power of a diffraction grating.

**[REPEATED: Identical to 2081 Q5]**

### **B. Long Answer Questions (LAQs)**

\*\*\*2082 Mid-Term Q6 (10):\*\*\*

- **(a) [7 Marks]:** Explain how coherent sources be obtained in practice? Define resolving power of grating and deduce its relation. [3+4] **[REPEATED: Similar to 2081 Q5, 2079 Q5]**
- **(b) [3 Marks]:** How many orders will be visible if the wave length of light be 5644 Å and the number of lines per inch on the transmission grating are 15000.

**2081 Q9(a) (10):**

What are the essential conditions for having Fraunhofer diffraction. Discuss Fraunhofer diffraction pattern due to single slit and determine the width of central maximum. [2+8]

**[REPEATED: Similar to 2079 Q11(a)]**

**2081 Mid-Term Q6 (16):**

- **(a) [11 Marks]:** What is a zone plate? Give its theory. Show that a zone plate has multiple foci. [1+3+7] **[REPEATED: Similar to 2080 Q9(a)]**
- **(b) [5 Marks]:** A screen is placed 2 m away from a narrow slit which is illuminated with light of wavelength 6000 Å. If the first minimum lies 5 mm on either side of central maximum. Calculate the slit width.

**2080 Q9 (15):**

- **(a) [10 Marks]:** What is zone plate? Show that how zone plate acts as converging lens of multiple foci. [2+8] **[REPEATED: Similar to 2081 Mid-Term Q6(a)]**
- **(b) [5 Marks]:** Calculate the highest order spectrum which may be seen with monochromatic light of wavelength 6000 Å by means of a diffraction grating with 5000 lines/cm.

**2079 Q11(a) (10):**

Differentiate between Fresnel and Fraunhofer diffraction. Explain Fraunhofer diffraction of light through single slit. **[REPEATED: Similar to 2081 Q9(a), 2082 Mid-Term Q4]**

### ***C. Trend Analysis & Focal Areas***

- ***Question Types & Styles:*** This chapter has the highest theoretical weight in wave optics. Grating resolving power derivations and Zone Plate multiple foci proofs are standard. Numericals focus on grating orders and slit widths.
- ***Direct Repetitions:***

- *Zone Plate multiple foci and focal length theory in 2080, 2081, and 2081 Mid-Term.*
- *Resolving power of grating derivation in 2079, 2081, and 2082 Mid-Term.*
- **Key High-Priority Topics:**
  - *Zone plate theory (deriving focal length and proving multiple foci).*
  - *Fraunhofer single-slit diffraction (deriving central maximum width).*
  - *Resolving power of a diffraction grating.*

## **Unit 6: Wave Optics: Polarization — [7% - 12%] of Total Marks**

### **A. Short Answer Questions (SAQs)**

\*\*\*2080 Q2(a) (4):\*\*\*

What is polarization? Differentiate between quarter wave plate and half wave plate. [1+3]

\*\*\*2080 Q6(a) (4):\*\*\*

Discuss the double refraction of light.

\*\*\*2079 Q2 (8):\*\*\*

What are the optical devices require producing circularly polarized light from unpolarized light? Explain how they are used to produce circularly polarized light.

### **B. Long Answer Questions (LAQs)**

\*\*\*2081 Q10 (15):\*\*\*

- **(a) [10 Marks]:** Define elliptically polarized lights. Explain with necessary theory, how to obtain and detect elliptically polarized light. [2+8]
- **(b) [5 Marks]:** Calculate the specific rotation if the plane of polarization is turned through  $26.4^\circ$  on traversing 20 cm length of 20% sugar solution.

**2079 Q9(b) (5):**

Plane polarized light passes through a calcite plate with its optic axis parallel to the faces. Calculate the least thickness of the plate for which the emergent beam will be plane polarized. Given  $\mu_o = 1.6584$  and  $\mu_e = 1.486$  and wavelength of light is  $5000 \text{ \AA}$ .

### **C. Trend Analysis & Focal Areas**

- **Question Types & Styles:** Dominated by advanced polarimetry. Compulsory theoretical questions on producing/detecting circularly and elliptically polarized light. Specific rotation numericals are heavily standard.
  - **Direct Repetitions:**
    - Production of circularly/elliptically polarized light using wave plates.
  - **Key High-Priority Topics:**
    - Quarter-wave and Half-wave plate theory.
    - Methods to produce and detect elliptically/circularly polarized light.
    - Specific rotation formula and calculation.
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## **Unit 7: Laser & Fiber Optics — [11% - 15%] of Total Marks**

### **A. Short Answer Questions (SAQs)**

\*\*\*2081 Q1 (8):\*\*\*

Define optical fiber and write down its application. Derive relation for numerical aperture. [4+4] [REPEATED: Similar to 2080 Q11(b), 2079 Q11(b)]

\*\*\*2081 Q6 (8):\*\*\*

What is laser? Write down and explain key characteristics of Laser. [2+6] [REPEATED: Similar to 2080 Q2(b)]

\*\*\*2080 Q2(b) (4):\*\*\*

Write the features of LASER? Write the differences between spontaneous emission and stimulated emission. [1+3] [REPEATED: Similar to 2081 SAQ6]

\*\*\*2079 Q4 (8):\*\*\*

Describe the working of Ruby laser.

### **B. Long Answer Questions (LAQs)**

\*\*\*2080 Q11 (15):\*\*\*

- (a) [5 Marks]: Describe working principle of He-Ne gas laser.
- (b) [5 Marks]: Derive the formula for acceptance angle for optical fiber. [REPEATED: Similar to 2081 Q1]

- **(c) [5 Marks]:** In an optical fiber, the core material has refractive index 1.6 and refractive index of clad material is 1.4. Calculate the values of critical angle and angle of acceptance.

**2079 Q11(b) (5):**

Calculate the numerical aperture and acceptance angle of an optical fibre from the following data:  $\mu_1$  (core) = 1.55 and  $\mu_2$  (clad) = 1.50.

**C. Trend Analysis & Focal Areas**

- **Question Types & Styles:** *Very high scoring. Usually splits into optical fiber parameters (acceptance angle, numerical aperture) and laser mechanisms (Ruby and He-Ne gas laser).*
  - **Direct Repetitions:**
    - *Derivation of Numerical Aperture and Acceptance Angle is almost guaranteed in every cycle (2079, 2080, 2081).*
  - **Key High-Priority Topics:**
    - *Derivation of Acceptance Angle ( $i_a$ ) and Numerical Aperture ( $NA = \sqrt{\mu_1^2 - \mu_2^2}$ ).*
    - *Difference between spontaneous and stimulated emission.*
    - *Construction and working of He-Ne and Ruby lasers.*
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## SUBJECT PRIORITY PLANNER

| Rank | Unit No. | Unit Name                  | Exam Priority Class | Historical Mark Percentage | Core Focus Areas to Pass/Score                                                                                |
|------|----------|----------------------------|---------------------|----------------------------|---------------------------------------------------------------------------------------------------------------|
| 1    | Unit 1   | Elasticity                 | Highest             | 16% - 27%                  | $K = Y / [3(1 - 2\sigma)]$ proof, Cantilever deflection, wire twist torque, solid vs hollow cylinder torsion. |
| 2    | Unit 5   | Wave Optics: Diffraction   | Highest             | 15% - 21%                  | Zone plate focal length proof, Fraunhofer single slit diffraction derivation, Grating resolving power.        |
| 3    | Unit 2   | Fluid Dynamics & Viscosity | High                | 15% - 19%                  | Bernoulli's derivation, Poiseuille's formula derivation, falling water drop terminal velocity calculation.    |
| 4    | Unit 4   | Wave Optics: Interference  | High                | 11% - 19%                  | Fresnel's Biprism wavelength determination, wedge-shaped thin film fringe width, Michelson setup.             |
| 5    | Unit 7   | Laser & Fiber Optics       | High                | 11% - 15%                  | NA and Acceptance angle derivation/numericals, He-Ne & Ruby laser working, Spontaneous vs Stimulated.         |
| 6    | Unit 6   | Wave Optics: Polarization  | Medium              | 7% - 12%                   | Circularly/Elliptically polarized light production/detection, quarter/half wave plates, Specific rotation.    |
| 7    | Unit 3   | Surface Tension            | Medium-Low          | 7% - 8%                    | Excess pressure inside drop derivation, Mercury droplet spraying work done numerical.                         |